



Anti-interference small target tracking from infrared dual waveband imagery

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ABSTRACT

For infrared small target detection and tracking, the challenge mainly contains dim targets, cluttered background, and infrared decoys. In the face of the above difficulties, traditional tracking algorithms focus on utilizing information of a single waveband is difficult to achieve good performance. In this paper, a tracking algorithm, which combines the Kernelized Correlation Filter (KCF) with a detection model, is presented to cope with these challenges on dual waveband imagery. Firstly, the Side Window Box Filter (SWBF) is utilized to suppress the background edge in the wide-band image, which contains much more detail than that in narrow-band image. Therefore, several filters are obtained in the first wide-band image processed by the SWBF. Subsequently, fused features for the KCF algorithm can be computed via convolution operation. In order to solve the migration problem referring to decoys, a detection model based on multi-frame association and image correlation is introduced to refine the target. Finally, experimental results demonstrate that the presented algorithm outperforms several relative algorithms in the accuracy and robustness. Besides, the multi-source information leads to improve the performance of tracking methods.

1. Introduction

The main task of IR target tracking is to constantly estimate the target position in successive frames [1]. Generally, the target in Infrared Radiation (IR) images is small in size, due to the fact of that the target is far away from the detection system. Besides, the object submerges in cluttered background [2–4], such as sea and sky. Furthermore, the performance can be reduced by the background edge that is passive and the active decoy. Therefore, it is considered as a difficult task [5–8] to tracking a small target.

Recently, IR tracking has been developed rapidly, including Template Matching Tracking (TMT) [9,10] algorithms, Mean Shift (MS) [11] algorithms, Correlation Filters (CF) algorithms [12–15], Deep Learning (DL) [16–19] algorithms, and some other feature extraction based algorithms [2,20]. The TMT match the target via the gray feature [9]. In the MS, a kernel histogram is utilized to obtain target position in the next frame. Yet, cluttered background makes the feature less discriminative and results in the problem of cumulative error. The Kernelized

Correlation Filters (KCF) [21,22] algorithm is widely applied in object tracking, due to its efficiency enabled by Fast Fourier Transform (FFT). This algorithm produces effective filters to obtain the structural information of the object of interest. Several improvements focus on obtaining more discriminative multichannel features, such as HOG. In [23], a two-layer Convolutional Neural Network (CNN) is presented to achieve robust representation. This effective feature is proved to be successful in the field of IR small target tracking [24], named GIF-CNN. Moreover, the CF is combined with deep feature although it may be time-consuming. Nevertheless, it is difficult for small target to train a suitable network, because of its easily overfitting.

Subsequently, target detection helps to refine the object of interest after that the IR system suffers from the problem of tracking migration [25,26]. In general, IR detection algorithms of small targets can be divided into three types, the image filtering based algorithm [27–29], the track association based algorithm [30,31] and eigenspace decomposition based algorithms [25,8,32–34]. The first approach utilizes filtering tools (Median filter, wavelet filter [28], Guided Image Filter

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(GIF) [27] and Side Window Box Filter (SWBF) [29]) to estimate the background, and the target information can be enhanced via background suppression. Besides, multiple frame filtering algorithms (Three frame filter [30], mean filter with updata mechanism [31] and so on) obtain more accurate result that benefits from temporal information of images. However, these filtering algorithms are easily influenced by the background clutter or edge, with similar local high-frequency information. The second type is achieved by continuity and trajectory consistency of moving targets in multiple frames. Afterwards, real targets can be filtrated from candidate targets. Specifically, Moving Pipeline Filtering (MPF) [35,36] is a classical algorithm that has been widely applied in the field of IR target detection. The third methods utilize the strategy of matrix decomposition to obtain the feature representation, such as wavelet transform [37], Principal Component Analysis (PCA) [8] and sparse representation [33,8,38]. Gao proposed a Infrared Patch Image (IPI) model [25] for small target detection, which is transformed into an optimization problem of recovering low-rank and sparse matrices. Yet, it is sensitive to IR background clutter.

The above algorithms focus on detection and tracking over images of one single waveband, which contain less detail than that of multiple source information. In this paper, we run the target tracking algorithm using the Dual Waveband Image (DWI), which is shown in Fig. 1. The DWI contain much more detail than that in one band image. Specifically, different spectral response of an object and the same spectral response of different objects are different. For the Wide Waveband (WW) with a longer wavelength, the response of both the target and the background can be strong. Besides, the Narrow Waveband (NW) with a shorter wavelength has a strong ability to restrain background. In order to track a small target under complex environment, this paper presents an IR small target tracking algorithm via a collaborative framework that integrates the KCF tracker based on fused features with the multi-frame detection based on image correlation. The detection model can improve the anti-interference performance. In the tracking model, the convolutional feature of the DWI can be more discriminative. Generally speaking, the paper provides researches with a small target tracking method via multi-source information.

The remainder of this paper is organized as follows. Section 2 gives the detailed description on the image filtering processing. In Section 3, the collaborative framework consisting of the target tracking model and the detection model is introduced. In Section 4, qualitative and quantitative results are introduced to verify the performance of the presented algorithm. Finally, conclusions are drawn in Section 5. The paper is concluded in Section 6.

2. Image processing

Sub Window Box Filtering (SWBF) [29] does well in the field of image denoising and can also preserve edge in a efficient way (Pixels to be processed are at the edge of the filtering window). Given a pixel (x, y) , eight sub-windows $N_i (i \in [1, 8])$ are defined as following

$$N_1(x, y) = \{(u, v) | x - r \leq u \leq x, y \leq v \leq y + r\} \quad (1)$$

$$N_2(x, y) = \{(u, v) | x \leq u \leq x + r, y \leq v \leq y + r\} \quad (2)$$

$$N_3(x, y) = \{(u, v) | x \leq u \leq x + r, y - r \leq v \leq y\} \quad (3)$$

$$N_4(x, y) = \{(u, v) | x - r \leq u \leq x, y - r \leq v \leq y\} \quad (4)$$

$$N_5(x, y) = \{(u, v) | x - r \leq u \leq x, |v - y| \leq r\} \quad (5)$$

$$N_6(x, y) = \{(u, v) | x \leq u \leq x + r, |v - y| \leq r\} \quad (6)$$

$$N_7(x, y) = \{(u, v) | |u - x| \leq r, y \leq v \leq y + r\} \quad (7)$$

$$N_8(x, y) = \{(u, v) | |u - x| \leq r, y - r \leq v \leq y\} \quad (8)$$

where r denotes the filtering radius. To automatically choose one window, the problem is represented as

$$U^{t+1} = \arg \min_U \left(\min_{i \in [1, 8]} |k_i * U - U^t| dx dy \right) \quad (9)$$

where U^t is the filtered result after $(t-1)$ -th iterations. Besides, the i -th kernel k_i is described in box filter as following

$$k_1(u, v) = \begin{cases} (r+1)^{-2} & (u, v) \in N_1 \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

$$k_5(u, v) = \begin{cases} 1/((r+1)(2r+1)) & (u, v) \in N_5 \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

where (k_2, k_3, k_4) and (k_6, k_7, k_8) are similar to k_1, k_5 , respectively.

In this paper, the WW image contains much background information, which can degrade the performance of most trackers using FFT technology. Thus, we run the SWBF algorithm on this waveband to reduce the influence of the edge in the background. Several experimental results are shown in Fig. 2.

As shown in Fig. 2(a), the location of the small target are close to that of coastal zone (i.e., background edge), which makes the tracking task fail. However, most detail, including the corner and edge, is preserved in the filtering result (red line in Fig. 2(a) and (b)). The target image is



(a) Mean: 4.9um, Wavelength: 10um



(b) Mean: 5.8um, Wavelength: 6um

Fig. 1. Dual Waveband Images of sky1 sequence.

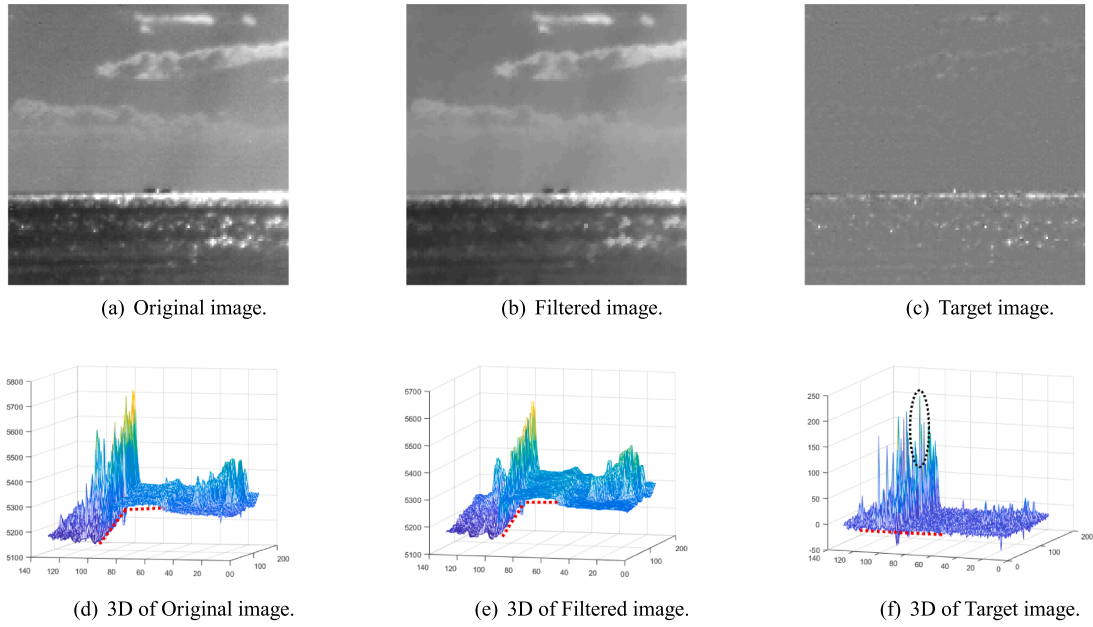


Fig. 2. The result of SWBF on sea1 images.

represented as the difference between the filtered image and the original image. In Fig. 2(c), the background edge is suppressed and target information is enhanced (Black ellipse in Fig. 2(f)).

3. Tracking on IR dual waveband imagery

Each waveband has its advantage in feature representation. In this section, we firstly briefly describe the KCF tracker [21], and then extend it to use the effective information of the two wavebands. Finally, the detection model is introduced to improve the anti-interference performance.

3.1. Correlation filter tracker via fused feature

The key is to train a classifier through a ridge regression model, which is represented as

$$\min_w ((w^T x - y)^2 + \lambda \|w\|^2) \quad (12)$$

where y, λ and w represent the regression result, parameter and coefficient, respectively.

The KCF densely samples a circulant matrix $X = C(x)$, where $C(x)$ denotes the circulant operation based on the first row x (i.e., base sample). Improved via the kernel trick, coefficient w is mapped to a high-dimensional feature space, i.e., $w = \alpha^T \varphi(x)$, where $\varphi(\cdot)$ means the mapping function. Then, the new coefficient α can be represented as

$$F\alpha = ((Fk^{xx}) + \lambda)^{-1} (Fy) \quad (13)$$

where F represents the FFT matrix, k^{xx} can be described as

$$k^{xx} = \exp\left(-\frac{(\|x\|^2 + \|z\|^2)}{\sigma^2}\right) - 2F^{-1}\left(\sum_d (Fx_d) \otimes Fz_d\right) \quad (14)$$

where the multichannel feature x is represented as $[x_1, x_2, \dots, x_d]$, x_d is the d -th target feature. More information can be find in [21].

Subsequently, the image patch z in the current frame is treated as the base sample of a testset. Finally, the response map is

$$f(z) = F^{-1}((Fk^{xx}) \otimes (F\alpha)) \quad (15)$$

In [23], some filters are sampled from the target template in the

current frame, in order to obtain convolutional features without training. Beside, this technology has been applied in tracking small targets and proved to be efficient [24]. The WW image contains more information, thus, filters can be obtained to be the prior in the first frame, which has been processed by the SWBF. Further, multichannel features can be obtained by the convolution operation between the these filters and the NW image. The experimental result referring to one filter is shown in Fig. 3.

As shown in Fig. 3, features of three targets have a similar pattern even with different appearance, which proves that local filters can extract useful information. For background templates that is far away from the target, the features undergo larger difference. In short, this filter can extract local feature for small targets. Therefore, the convolutional features can be utilized as multichannel features in KCF. Specifically, the WW has been processed by SWBF and several filters are extracted at the first frame.

3.2. Anti-interference method based on track association and image correlation

For most tracking algorithm such as KCF [39] and DL based tracking [40], the cumulative error becomes larger and larger. Thus, background information is mixed into the updated target template. Generally, traditional tracking algorithms do not utilized detection method to re-search targets, which is because of two reasons. One is that the global searching task is time-consuming, and another is that small target detection can be easily influenced by the background clutter or edge. Therefore, we introduce an efficient algorithm based on track association and image correlation.

3.2.1. Track association method

The Moving Pipeline Filter (MPF) [35], whose key is to establish a spatial pipeline centered on the target, has been applied to estimate the object trajectory. Due to the fact of the target's continuity in the sequence, the target occurs in the neighbourhood centered on the target (x, y) in the first frame of the pipeline. The neighborhood size and the number of frames that need to be processed are represented by the diameter D and length N of the pipeline, respectively. The main step is described as follow.

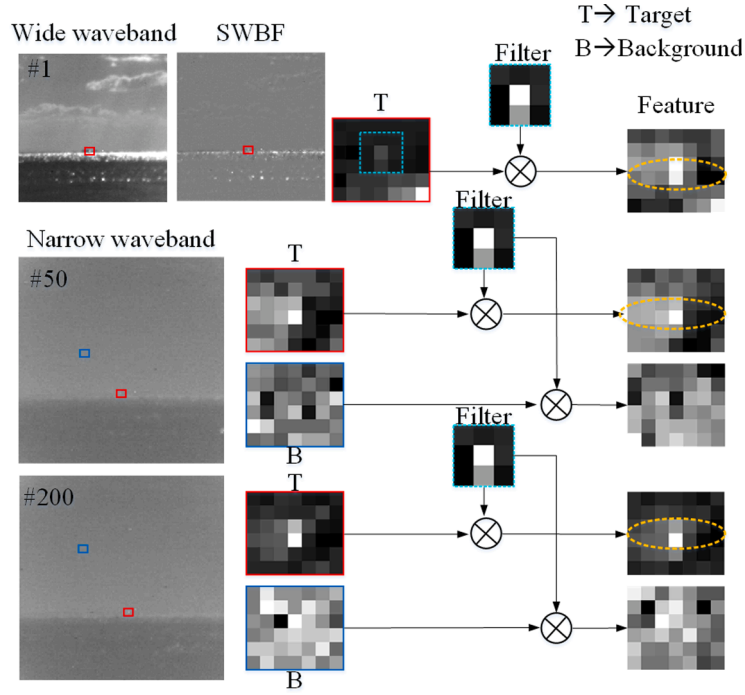


Fig. 3. The convolutional feature of DWI without training.

- (1) Initialization. Setting the pipeline length N and the diameter D .
- (2) Determine the target P_i in the first frame using the top hat approach.
- (3) If a new target occurs in the neighbourhood of P_i in the next frame, the frame counter K is increased by one. Besides, the target location is updated.
- (4) Once N frames are all processed, the counter value is obtained. The candidate whose counter value is no less than K_1 is identified as a target. Otherwise, it should be seen as a false target and deleted from the pipeline.
- (5) Update the image in the pipeline and continue to process all images. For the current pipeline, remove the first frame, move the other images forward by one frame and put the new frame into the end of this pipeline.

Therefore, in the trajectory of targets, the target location in the current image is the detecting result of the previous image (Shown in Fig. 4). Generally, the pipeline length N equals to 5 and the threshold value K_1 is 4.

3.2.2. Image correlation

Here, the radiation energy of decoys is much higher than that of

small targets, and its size is times larger than the target size. Therefore, the detection model is enabled after the frame that the target's intensity suddenly increases. As shown in Fig. 4(b), the MPF outputs several results because of decoys, which can make tracking fail easily. Based on the gray feature, we use the image correlation to determine the true target. Give image X and Y_i , the correlation value Z_i can be represented as

$$Z_i = \text{Corr}(X, Y_i) / (\sigma_X \sigma_{Y_i}) \quad (16)$$

where Corr denotes the correlation function, and σ is the standard deviation.

At the frame that the decoy just occurs, Y_i represents i -th detecting results of the MPF algorithm, and X denotes the target template in previous frame. The experimental result is shown in Fig. 5. Specifically, larger Z means the candidate target is more likely to be a target.

The flowchart of the presented KCFD with Detection (KCFD) algorithm is shown in Fig. 6.

4. Experimental analysis

In this section, the KCFD algorithm is tested over six challenging DWI



Fig. 4. Object trajectory obtained via the MPF algorithm.

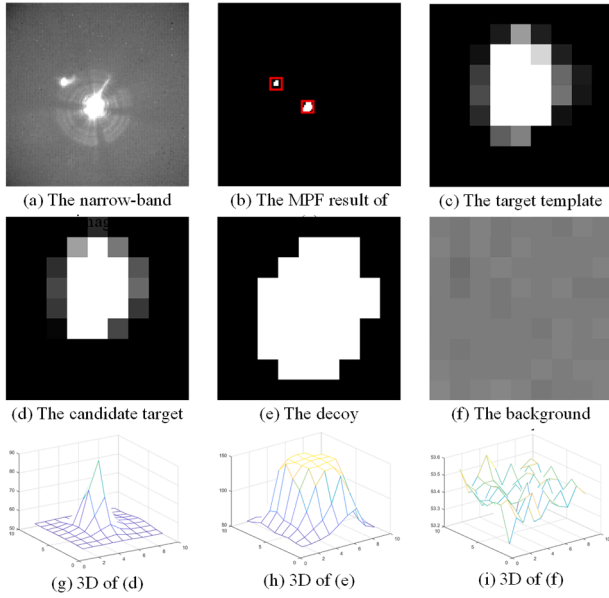


Fig. 5. Image correlation between the target template and the outputs of the MPF algorithm. $Z_{cd} = 0.8$, $Z_{ce} = 0.47$ and $Z_{cf} = -0.04$, where Z_{cd} denotes the corrccoef between (c) and (d).

sequences, which is collected via two IR cameras with the same field-of-view but different wavebands. Six sequences are named *sky1*, *sky2*, *sea1*, *sea2*, *decoy1* and *decoy2*, whose details are described in Table 1.

4.1. Experimental setup

All experiments are implemented using MATLAB 2017a on a PC with 32 GB RAM and NVIDIA GeForce GTX 1070. Comparisons consist of the MPF [35], the IPI [25], the DLT [40], and several of our previous studies (the Improved KCF (IKCF) [39], the GIF-CNN [24] and SDC-CNGM [41]). Specifically, these algorithms are performed on the two waveband sequences, respectively. Further, the parameters are set as follow.

At the beginning of the proposed KCFD algorithm, parameters of the KCF are automatically set as fix values. Besides, In order to extract fused features for the KCF, both the size and the number of the filter are set as 3. In the SWBF, the radius r is 1 and the iteration t is fixed as 10. In the

MPF, the pipeline length N equals to 5 and the threshold value K_1 is 4. Qualitative results of four tracking algorithms are shown in Figs. 7–12. We compare the KCFD with the other algorithms over three situations, i. e., sea background, sky background and decoy interference.

Subsequently, parameters in the MPF algorithm and the IKCF algorithm are same as those in the KCFD. For the IPI algorithm, all parameters are not changed, and adaptive threshold segmentation is utilized to obtain the target state. In the DLT algorithm, the size value referring to the input image is fixed as an appropriate value, and the particle number is reduced to a low value for high efficiency. Besides, the DLT with the same parameters performs on the Graphics Processing Unit (GPU) and the Central Processing Unit (CPU), respectively. Furthermore, for the GIF-CNN and the SDC-CNGM algorithms, nothing has been done in their parameters, except for a few minor modifications in image size or particle number.

4.2. Qualitative comparison

4.2.1. Sea background

In *sea1* sequences, the dim and small target on boats is surrounded by the background clutter near the sea-horizon. In the WW image, the IPI, the IKCF and the MPF fail to obtain the target position at frame 148 and frame 198, which is shown in Figs. 7(c)-(d). The DLT algorithm undergoes migration at frame 50 and frame 148 (Figs. 7(b)-(c)). Besides, in the NW image, the target intensity is weak. The MPF and IPI obtain false targets, the IKCF and the DLT fail to track the target after frame 50 (Fig. 7(f)). Benefiting from the SWBF and the fused feature, the KCFD can obtain more discriminative target information, and the tracking

Table 1
Overview of test IR sequences.

Sequence	Frame	Target size	Description
<i>sea1</i>	200	3×3	A lamp with sea background clutter
<i>sea2</i>	150	3×3	A lamp with sea background clutter and the mountain
<i>sky1</i>	200	4×4	A plane with sky background clutter and partial occlusion
<i>sky2</i>	300	5×5	A plane with sky background clutter
<i>decoy1</i>	150	9×9	A plane with point decoy
<i>decoy2</i>	150	9×9	A plane with non-point decoy

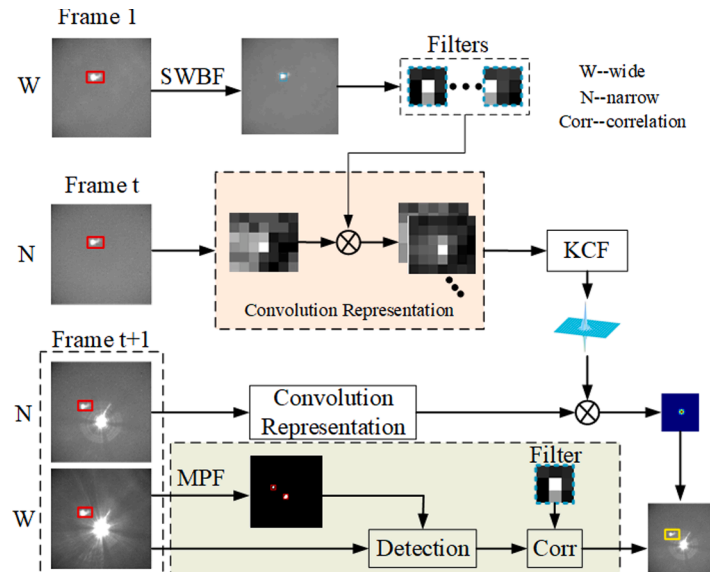


Fig. 6. The flowchart of the presented algorithm.

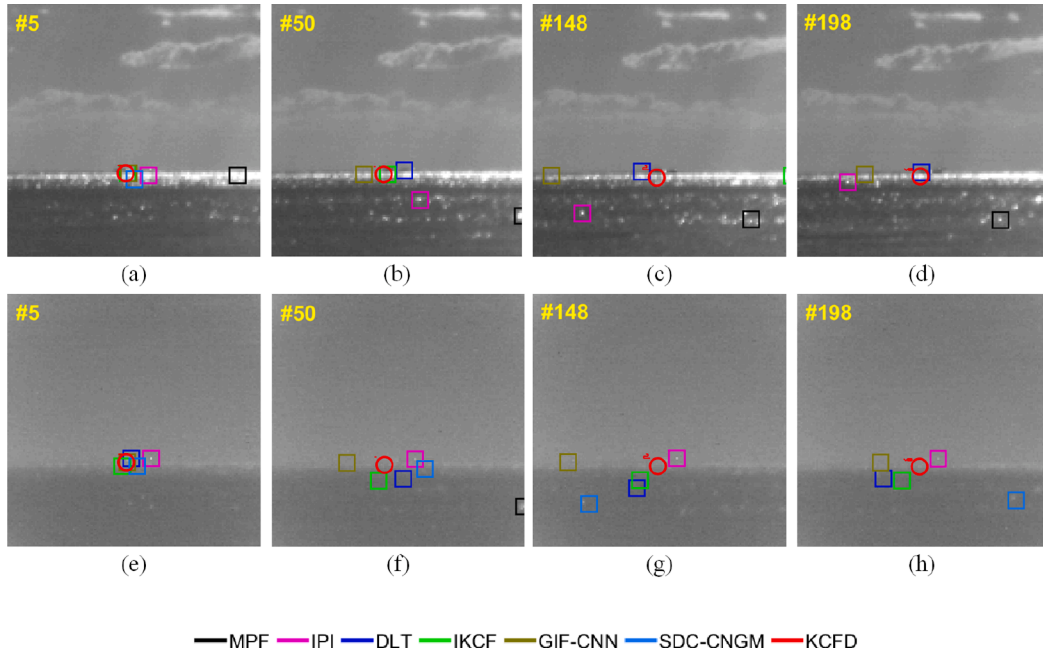


Fig. 7. Qualitative results over *sea1* sequence, (a)-(d) and (e)-(h) represent results of WW and NW images, respectively. (Best viewed on color display).

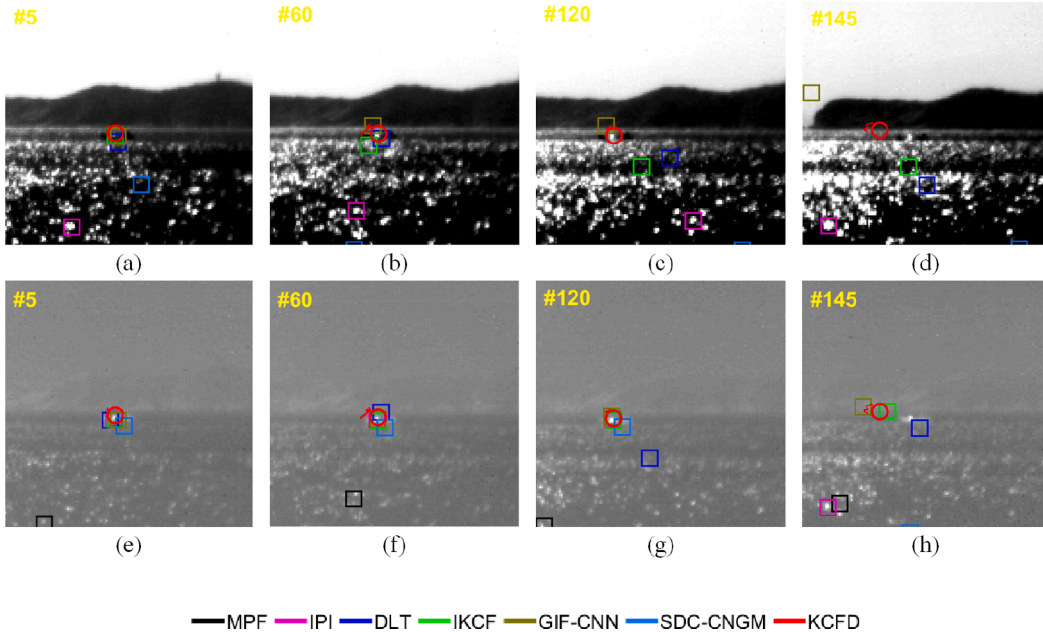


Fig. 8. Qualitative results over *sea2* sequence.

result is more satisfactory.

In *sea2* sequences, the sea background undergoes strong radiation and the target is under the mountain. The MPF algorithm is sensitive to the background edge, for the reason that the background clutter forms a pipeline, which is detected as a target easily. The IPI mistakes noise points as targets easily. The DLT and GIF-CNN are influenced by the strong clutter around the target at frame 120 (i.e., Fig. 8(c)). Besides, in the NW image, the IKCF, SDC-CNGM and the proposed algorithm can obtain the target position at frame 5, 60 and 120, as shown in Figs. 8(e)-(g). However, all algorithms fail to obtain the target position at frame 145 (Fig. 8(h)), because of camera shake.

4.2.2. Sky background

In *sky1* sequences, the target is partially occluded by the tower at frame 160. As shown in Fig. 16(c)-(d), the candidate template in the DLT algorithm is updated with gray feature of the tower, which results in tracking failure. It is the same for the IKCF algorithm and the GIF-CNN. Both the MPF, IPI and KCFD take the advantage of the detection model, thus the target can be refind at frame 190. In *sky2* sequences, the plane target submerges in the cloud at some point. Specifically, in the WW image (Figs. 9(a)-(d)), the DLT algorithm can not find the true target at frame 100 and 295. The tracking results of the IKCF algorithm and the GIF-CNN algorithm drift to the background at frame 100. Besides, in the NW image (Figs. 9(e)-(f)), the DLT and the IKCF suffer from a large migration at frame 295.

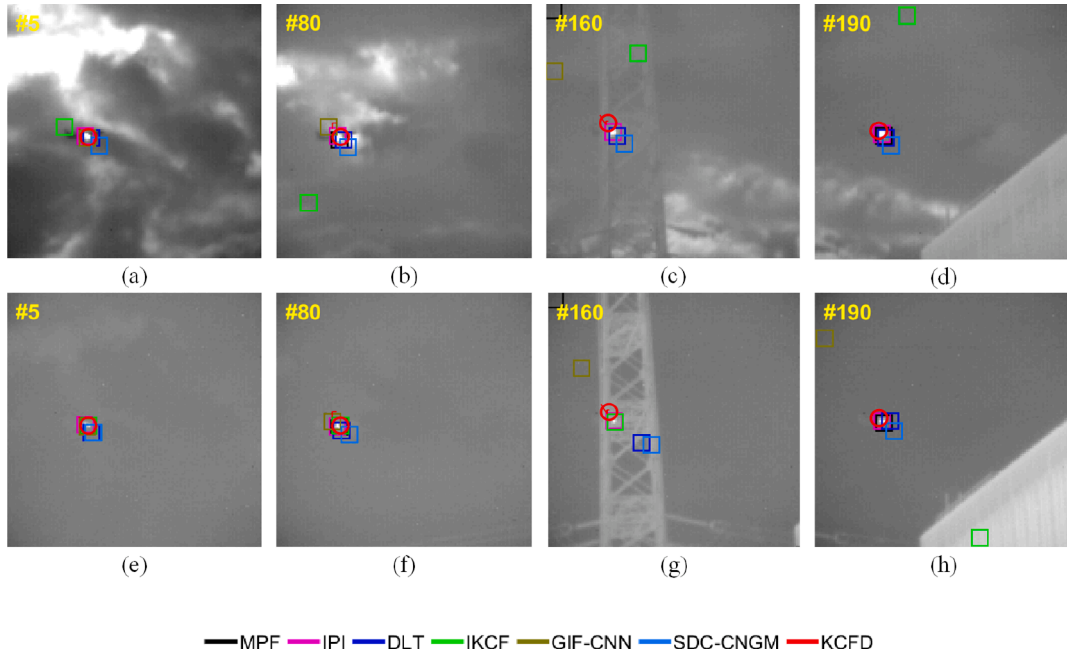


Fig. 9. Qualitative results over sky1 sequence.

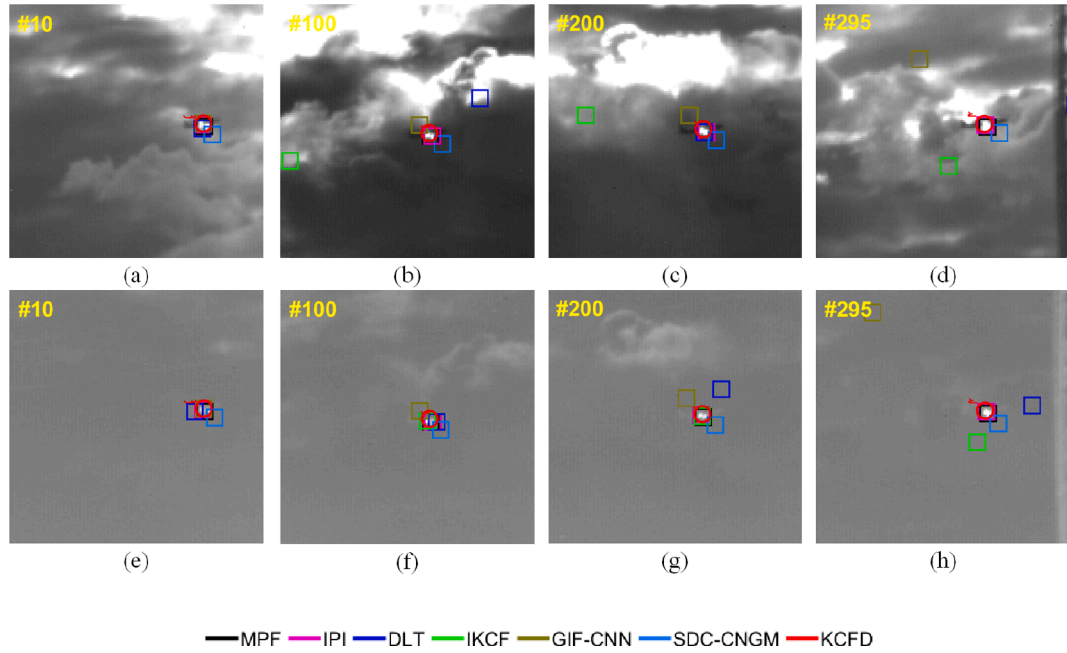


Fig. 10. Qualitative results over sky2 sequence.

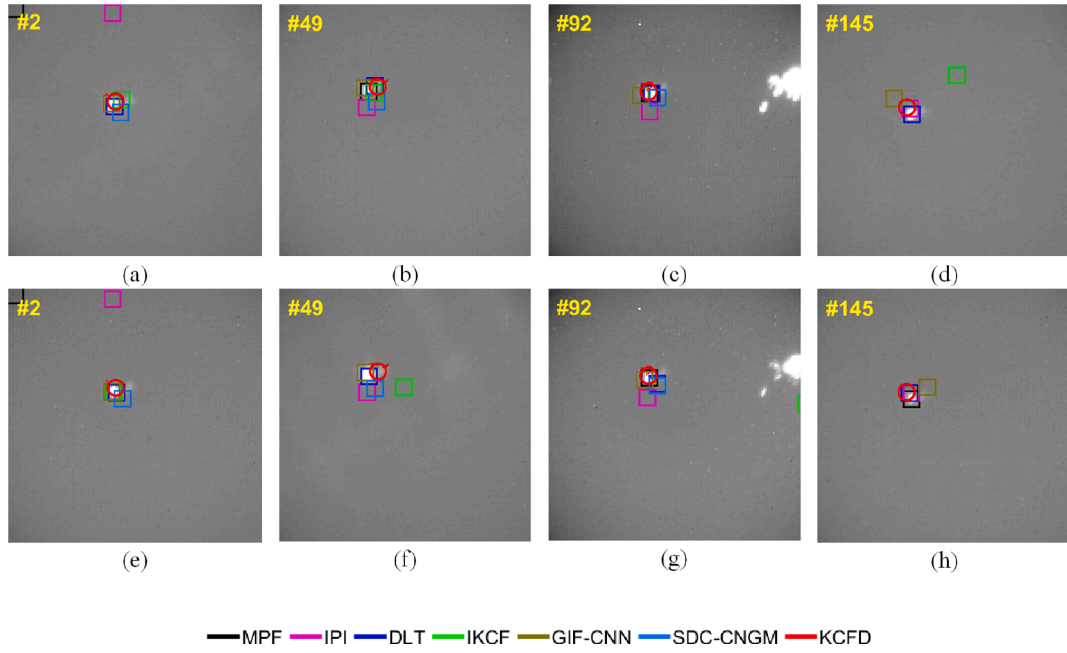
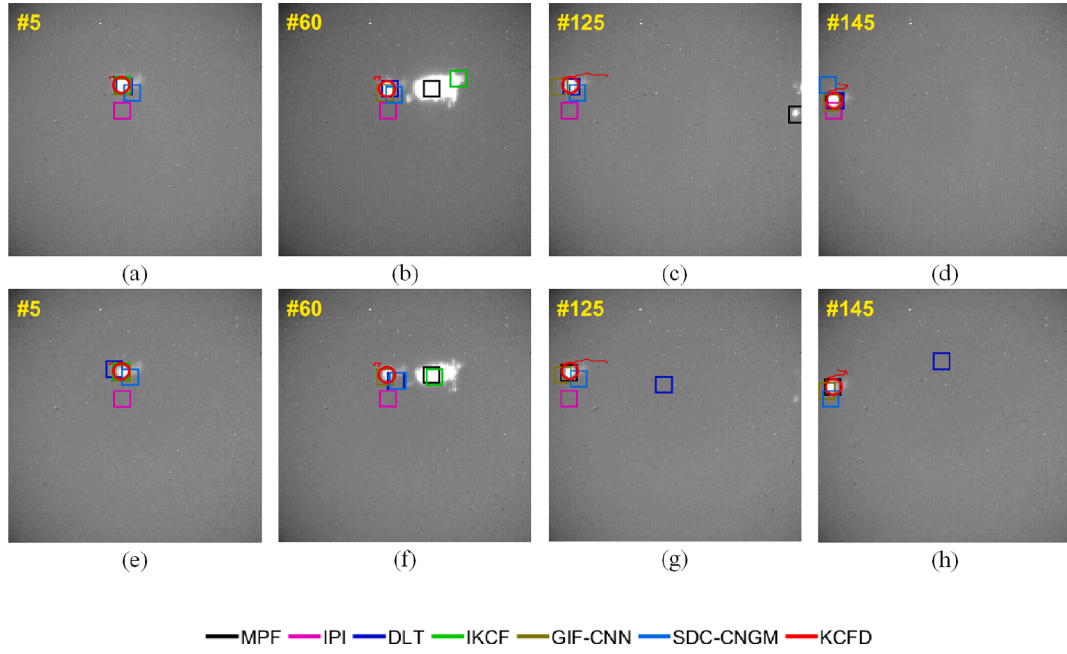
4.2.3. Decoys

In *decoy1* sequences, target detection is influenced by the decoy at some point. The decoy is strong in radiation and covers the target at the frame it occurs. The MPF mistakes the clutter as a target, which is shown in Fig. 11(a)–(e). Besides, the KCF based algorithm drifts to the background at frame 145 (Fig. 11(d)). Furthermore, The IPI, the DLT and the proposed KCFD obtain the target position accurately (frame 145). The *decoy2* sequence contains a decoy with large size. The DLT algorithm drifts to background only in NW images (Figs. 12(f)–(h)), for the reason that deep feature is overfitting for small target tracking. Besides, the IKCF and MPF trackers are easily influenced by the non-point object (i. e., at frame 60), and the problem is unrecoverable for the IKCF tracker.

4.3. Quantitative comparison

In this section, we utilize Center Location Error (CLE), Success Rate (SR) and Frame per Second (FPS) to evaluate the tracking performance [39]. The CLE is represented by the Euclidean distance E between the estimated and the real target. The second evaluation is defined as M/S , where S denotes the sequence length, and M denotes the length of successful tracking (i.e., $E < 8$). Then, comparison results are shown in Fig. 13 and Table 2.

In Fig. 13, log means logarithm and the line ' $y = \log(8)$ ' represents a threshold. As shown in Figs. 13(a)–(d), it is obvious that the MPF algorithm have a large fluctuation in the CLE curves, caused by the cluttered background. Besides, the DLT and IKCF drift to the background

Fig. 11. Qualitative results over *decoy1* sequence.Fig. 12. Qualitative results over *decoy2* sequence.

slowly once it loses the target position. Yet, the CLE of DLT on WW images is better than that on NW images. In Figs. 13(e)–(h), most trackers obtain accurate position of targets. Nevertheless, the CLE curves of IKCF tracker has a large fluctuation that can be seen in Fig. 13(e) and (g). The reason is that the background contains edge of cloud, which has the same characteristics in frequency-domain. Specially, the tracking results on NW images are satisfactory. Finally, in the four sequences that contain a decoy (Fig. 13(i)–(l)), the CLE curve referring to the IKCF algorithm increases after frame 50, which is same as that of the GIF-CNN and the SDC-CNGM in *decoy1* sequence. It is difficult for this algorithm to adapt to the situation that the target is occluded by another object. The MPF algorithm and the IPI algorithm get a large CLE value, due to the fact of the decoy's continuous moving.

As shown in Table 2, the SR results of the MPF over the *sea1* and *sea2* sequences are as low as 11, due to the fact that this tracker is sensitive to the environment. The SR values of IPI over the NW images are relatively high, which indicate that the IPI algorithm is sensitive to the background clutter. The tracking migration in the KCF may result in an unrecoverable detection (Fig. 13(b), (e) and (i)), and the lowest SR value over these three sequences is 5. The DLT algorithm owns a modest SR value (Up to 1 over the decoy sequence), yet it is time-consuming. To be specific, its FPS is 10 (GPU) with the particle number set as 200 and the efficiency can be lower as the number becomes higher. Besides, it can be influenced by the edge (Fig. 13) and the SR over the WW image of *sea1* decreases to 0.61. The GIF-CNN and the SDC-CNGM perform well over NW images, but lose targets easily if the target is severely occluded.

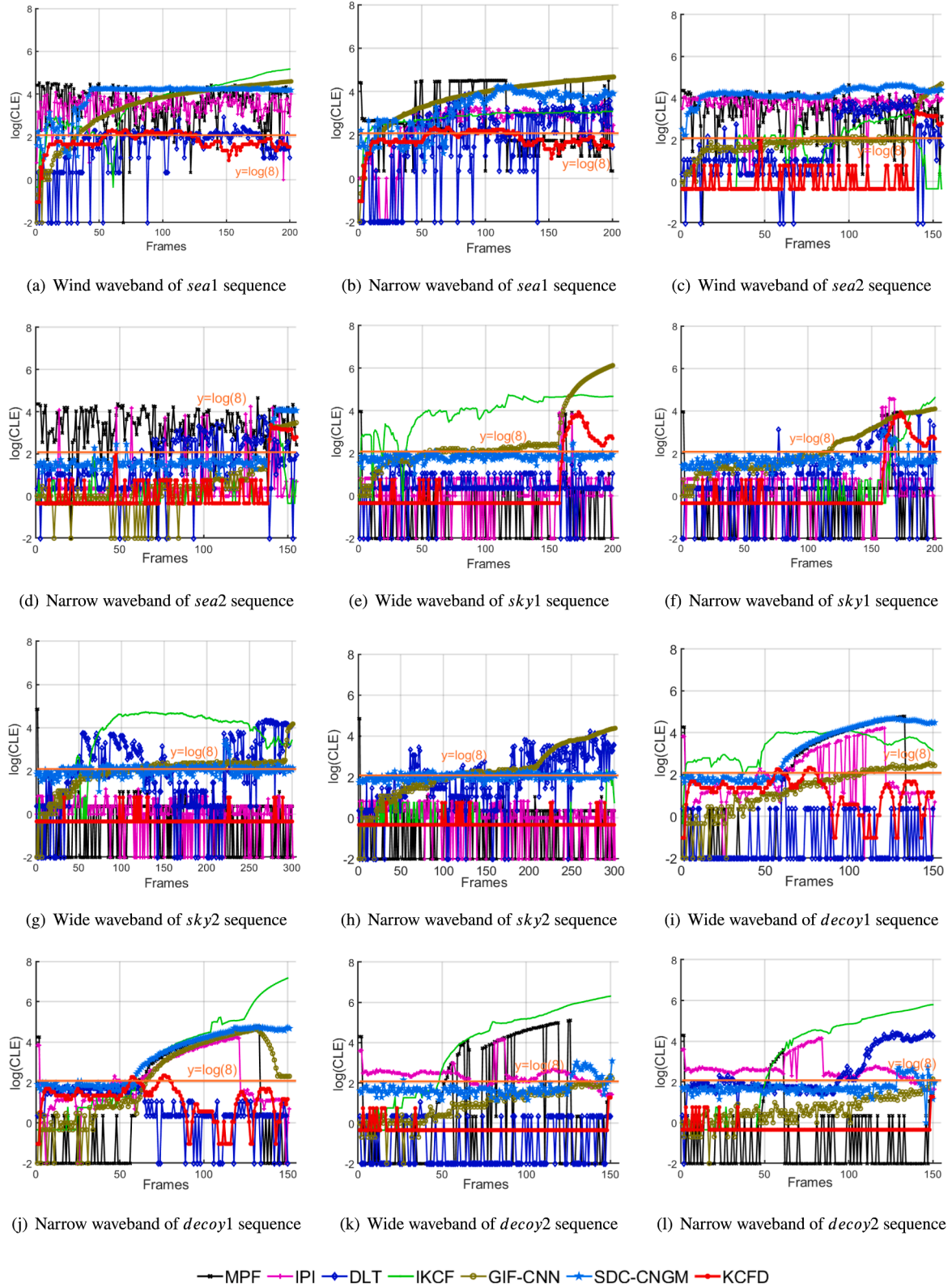


Fig. 13. The logarithm value of the center position error (CLE).

Moreover, from the average SR value in Table 2, it can be seen that the MPF performs well over the NW image, which contains less clutter. The situation is the same for the IPI, the IKCF and SDC-CNGM. Besides, the DLT and GIF-CNN focus on extracting a effective feature, thus, it obtains a more accurate SR value over the WW images.

Afterwards, we briefly discuss the complexity of the proposed method for small target tracking, which is shown in Fig. 6. The computational complexity of KCF [21] is as low as $O(n \log n)$, benefiting from FFT instead of expensive matrix algebra. As the SWBF implementation needs to perform calculations over multiple windows, its

Table 2SR (%) and FPS. The **Bold** and the *italic* fonts denote the best and the second best performance.

Sequence	sea1		sea2		sky1		sky2		decoy1		decoy2		Average SR			Average FPS
	WW	NW	WW	NW	WW	NW	WW	NW	WW	NW	WW	NW	WW	NW	ALL	
MPF	11	41	12	10	95	96	99	99	52	51	63	93	55	65	60	16
IPI	2	19	5	92	98	96	52	100	64	58	24	9	41	62	52	3
DLT	61	47	60	98	80	87	52	56	99	100	100	57	75	74	75	CPU1.6/GPU10
IKCF	14	4	61	92	5	82	19	99	3	39	32	33	22	58	40	75
GIF-CNN	16	8	85	92	42	58	42	41	71	43	97	98	59	57	58	20
SDC-CNGM	7	18	–	90	95	93	62	68	39	39	85	89	58	66	62	28
KCFD		68		92		96		100		92		100		91		30

computational cost is relatively higher. Nevertheless, the image size is 128×128 , and local area centered on the target location can be utilized to enhance the target. Besides, the filtering can be performed in advance. Finally, the detection model based on MPF is awakened once the decoy occurs. Based on the aforementioned analysis, the entire computational complexity of the proposed method mainly depends on that of the SWBF.

In short, the SWBF processing and the fused feature utilize sufficient target information even if the target has a low gray intensity. Besides, the detection model helps to refine the target. The CLE result of the presented KCFD over all test sequences is relatively satisfactory, proving that it performs better than related algorithms in the accuracy and robustness.

5. Conclusion

In this paper, we introduce a detection based correlation filter approach for small target tracking over infrared dual waveband images. The fused feature based on convolution provides discriminative information and can be used as multichannel features for the tracking framework. Besides, the multiple frame information and the image correlation help to find the target position continuously while facing the decoy. Most importantly, experimental results insist that dual waveband information can improve the tracking performance. This lays the foundation for developing future infrared tracking algorithms.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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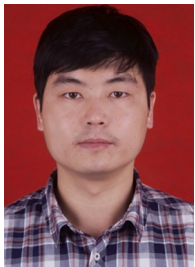
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